

by submarines. These are connected to land stations by underwater communication cables. Submarines hiding near such devices can stay in contact with their headquarters. Gertrude, which is an underwater telephone, can also be used to communicate with submarines.

Very low frequency (3kHz - 30kHz) transmission. This can penetrate seawater approximately 20 metres deep. A submarine at shallow depth can use these frequencies, while a vessel more deeply submerged might need a buoy equipped with an antenna on a long cable. The buoy rises to a few metres below the surface, and is small enough to remain undetected by enemy sonar and radar.

Extremely low frequency transmission. Electromagnetic waves (3Hz - 300Hz) can penetrate seawater to the depths of hundreds of metres, allowing signals to be sent to submarines at their operating depths.

Wireless technologies for diver automation

Seatooth LightRope is a smart, sub-sea wireless identification and location solution for divers and remotely-operated underwater vehicles (ROVs). It has a unique fibre-lighting system that offers a solution for challenging and hazardous sub-sea environments. It provides positive identification of structures, locations and umbilicals.

For example, when Seatooth LightRope is used by a diver, the ROV automatically switches on when the diver or ROV is five metres away. Its communication range through seawater is 500 metres and 3000 metres deep, with standard operating temperature of -0°C to +40°C.

Seatooth Wireless Communication and Control System (SWiCOM) is a secure and resilient underwater C4ISR solution for divers. It provides bi-direction support through water/air text communication up to nine metres. It supports remote sensor data exfiltration and communication with unmanned aerial vehicles.

SWiCOM can also help monitor the diver's core body temperature, wirelessly control jetboots and correct inertial navigation system.

Wireless personal area network (WPAN) supports a broad range of sensors and propulsion systems. It provides



Fig. 2: Wirelessly monitoring diver core body temperature (Credit: WFS Technologies)

a bandwidth of 1.2kbps - 2.4kbps user-selectable range, through seawater up to seven metres, with standard operating temperature of -10°C to +60°C. Data interfaces are RS232 and RS485, with storage capability of 16MB.

Wireless technologies for safety above the deep sea

Static pressure monitoring on offshore platforms is needed to monitor the pressure of the gas well remotely. It reduces maintenance cost and increases the safety of maintenance personnel. It also eliminates visits to the platforms for monitoring pressure. Also, it increases productivity due to early detection of pressure changes.

Remote leak detection for pipelines is needed to ensure safety, by monitoring oil leaks. It also lowers the burden on the environment by avoiding pollution. Facilities and personnel safety are also taken care of. Installation costs are reduced, since there is no need for cabling and anti-theft measures for cables.

Achieving the above with long-range wireless communication is difficult without repeaters, as it is not possible to place the repeaters between offshore platforms. There is a need to deal with height pattern variance, which is caused by tidal level changes.

Height pattern is the relationship between the height of antenna and signal strength. This pattern is generated by an interaction between radio waves arriving through different paths. Height pattern variation at offshore platforms is caused due to radio waves

through a direct path and reflected on sea surface. Height pattern at offshore platforms changes periodically, as relative height of the antenna from sea surface varies with tidal levels.

Ajoy Kumar, deputy general manager - field instruments business division, Yokogawa India Ltd, says, "Our solution for monitoring pressure is a plant-wide field wireless system that consists of YFGW710 (wireless integrated gateway), YFGW510 (wireless temperature transmitter), EJX530B (wireless pressure transmitter) and YFGW410 (wireless management station).

"Battery-operated pressure transmitters are used, which means there is no need for an electric power source at the well platform. Two access points are installed at different heights to prevent height pattern variance. A high-gain antenna and remote antenna cable are used for remote leak detection for onshore pipelines."

Other solutions for wireless communication for ships and submarines

For better connectivity and faster wireless device processing, dual-band transmission-balanced antennae combine ground-plane independence with high-radiation efficiency. This reduces engineering resources and costs needed to mitigate PCB ground-induced radiation. High radiation efficiency with a 34.90mm x 9mm strip antenna offers total efficiency values of at least 75 per cent minimum in 2.4GHz band and at least 70 per cent minimum in 5GHz band.

A base transceiver station (BTS) is designed for both point-to-point and point-to-multipoint network applications. The software engine allows the BTS to provide a user-friendly graphical user interface with installation tools for site survey, antenna alignment, delayed reboot and so on. It is compatible with the wireless network management system.

BTS Max by Maksat Technologies is a 5GHz point-to-multipoint product with superior performance for building long-distance links, super high power OFDM and 2 x 2 MIMO diversity 5.8GHz links. It is equipped with extreme output power up to 29dBm and 802.11n radio.

An integrated communications con-